

Program : Diploma in Biomedical Engineering	
Course Code : 6241A	Course Title: Advanced Medical Instrumentation and Prosthetics
Semester : 6	Credits: 4
Course Category : Program Elective	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester : 60

Course Objectives:

- To lay foundation on laser instrumentation, its generation and applications.
- To provide knowledge on various biomaterials and their application in human body.
- To give an account of the different orthotic, prosthetic and implants devices in biomedical fields.

Course Prerequisites:

Topic	Course code	Course name	Semester
Properties of light		Applied physics	1
Atomic structures and material properties		Applied chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the construction and working of biomedical lasers	16	Understanding
CO2	Describe the therapeutic and diagnostic applications of laser.	13	Understanding
CO3	Explain the various types of biomaterials, their properties and medical applications	13	Understanding
CO4	Describe different types of orthotic,prosthetic devices and various implants used in human body.	16	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the construction and working of biomedical lasers.		
M1.01	Explain the principle of operation of LASER	2	Understanding
M1.02	Illustrate the elements and mechanism of LASER system.	2	Understanding
M1.03	Discuss Q-switching and mode locking in LASERS.	2	Understanding
M1.04	Illustrate the construction and working of ruby laser, He-Ne laser, Nd-YAG, Argon ion, CO2 laser	10	Understanding
Contents: Laser working principle, generation and types LASER - Definition - properties - spontaneous and stimulated emission - population inversion - 3 level and 4 level laser - energy level diagram Basic elements and mechanism of laser generation - gain medium, Q-switching - concept of Q-switching, Mode locking - concept of mode locking, Ruby laser - construction, energy level diagram and working, He-Ne laser - construction, energy level diagram and working, Nd-YAG laser - construction, energy level diagram and working, Argon ion laser - construction, energy level diagram and working, CO2 laser - construction, energy level diagram and working :			
CO2	Describe the therapeutic and diagnostic applications of laser.		
M2.01	Sketch the schematic diagram and working of flow cytometry	3	Understanding
M2.02	Summarize the working optical bio sensors	2	Understanding

M2.03	State the working principle and application of optical coherence tomography with sketch	2	Understanding
M2.04	State the concept of photo sensitizer and mechanism of photodynamic therapy	3	Understanding
M2.05	List the application of lasers in- dermatology, neurosurgery and ophthalmology	3	Understanding
	Series Test – I	1	

Contents:

Flow cytometry, optical biosensors, photo dynamic therapy,application in various fields of medicine

Flow cytometry - principle and working, Optical bio sensors - principle and working

Optical coherence tomography- principle and working, photo sensitizer - principle and mechanism of action, Photodynamic therapy- principle and application, List the application of lasers in-dermatology, neurosurgery and in ophthalmology.

Case study: Prepare a presentation on LASIK surgery.

CO3	Describe the various types of biomaterials, their properties and medical applications.		
M3.01	Define biomaterials and biocompatibility, requirements of biomaterials, their classification	2	Understanding
M3.02	List the types of Polymers and explain their properties and uses	4	Understanding
M3.03	List the types of metals and metal alloys, ceramics and Composites explain their properties and uses	7	Understanding

Contents:

Classification of biomaterials- metals, ceramics, polymers, composites and their application

Define biomaterials - Requirements of biomaterials, biocompatibility,

Classification of biomaterials - Polymers - Metal and metal alloys - Ceramics - composites - properties, types and applications of each class

Polymers - Polymerization, Polyethylene, Biodegradable polymers, Blood compatible polymers, Bioactive polymers, Hydrogels - properties and uses

Metals and metal alloys - Stainless steel, Titanium and Titanium alloys, Cobalt based alloy Nitinol,

Dental metals

Ceramics - Types, properties and applications

Composites - Types, properties and applications

CO4	Describe about orthotic and prosthetic devices and various implants used in human body.		
M4.01	Differentiate between orthotic and prosthetic devices with application examples	1	Understanding
M4.02	Describe cardiovascular implants, their construction and application	5	Understanding
M4.03	Describe orthopedic implants, their construction and application	6	Understanding
M4.04	Explain dental implants, their structure and treatment modalities	3	Understanding
M4.05	Conduct a survey on total artificial heart.	1	Understanding
	Series Test – II	1	

Contents:

Cardiovascular implants, Orthopedic implants, dental implants

Prosthetic devices and Orthotic devices - definition and examples

Homograft and heterograft - definition, comparison, and examples

Cardiovascular implants: Vascular grafts, Artificial Heart Valves-types -tilting-disc, bileaflet and caged ball.

Orthopedic implants: Fracture fixation devices - External and internal fracture fixation devices - Joint replacements - Total Hip replacement (with schematic diagram)-, Total knee replacement (with schematic diagram), Bone regeneration with resorbable material

Dental implants: Dental implants - structure and treatment modalities

Case study: Conduct a survey on total artificial heart.

Text/ Reference:

T/R	Book Title/Author
R1	Sujatha v Butt <i>Biomaterials</i> – Alpha Science International, 2005
R2	DN Ghista Taylor& Francis <i>Biomechanics of medical devices-</i> , 1981
R3	R S Khandpur- <i>Handbook of Biomedical instrumentation-</i> McGraw Hill Education; Third edition (4 August 2014)

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/113104009/
2	https://www.ahajournals.org/doi/full/10.1161/CIRCULATIONAHA.108.778886
3	https://www.hipandknee.com/hip-surgery/about-hip-replacement/parts-materials
4	https://www.youtube.com/watch?v=WJ05XOJiaDY
5	https://www.youtube.com/watch?v=2Oswmij538Q